

AKINSOLA CLEMENT MORAKINYO

Bioinformatician | Cancer Genomics, NGS Analysis & AI Model Evaluation

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL PROFILE

Bioinformatician and Clinical Data Scientist with a strong quantitative research background and hands-on experience in NGS and cancer genomics. Experienced in WGS, WES, bulk RNA-seq and scRNA-seq analysis, with practical expertise in GATK4/Mutect2, SAMtools, DESeq2, Seurat and Nextflow/nf-core, and currently extending into spatial transcriptomics. Builds reproducible, containerised workflows and applies clinical research expertise to genomic and biomedical datasets.

TECHNICAL SKILLS

Core NGS & Variant Analysis: GATK4/Mutect2 (somatic calling, BQSR), BWA-MEM, SAMtools, bedtools, SnpEff, FastQC, Trimmomatic, BLAST

Transcriptomics & Single-cell: DESeq2, Seurat v5, Harmony, clusterProfiler/GSEA, Scanpy; spatial transcriptomics (in progress)

Workflows & Compute: Nextflow/nf-core, Docker, AWS EC2, HPC, Google Colab, Git/GitHub

Programming: Python (Biopython), R (Bioconductor), Bash/Linux, SQL

Data Types: FASTQ, BAM, VCF, count/expression and single-cell matrices, methylation arrays, MRI, clinical datasets

AI & Machine Learning: LLM and RAG applications, deep learning (3D U-Net), predictive modelling

Also Proficient: GWAS QC, COSMIC mutational signatures, MEGA phylogenetics, survival analysis (Kaplan–Meier, Cox), Streamlit, Power BI, REDCap, SPSS, Stata, Epi Info

PROFESSIONAL EXPERIENCE

Research Intern — Medical Artificial Intelligence Laboratory (MAI Lab)

2026 – Present

Lagos, Nigeria

- Curated brain MRI data for medical AI: organised multi-sequence scans and segmentation labels, checked image–label consistency and quality, and prepared standardised datasets for model training and evaluation.
- Built and deployed a BraTS-Africa 3D U-Net brain tumour segmentation model on African glioma MRI data (SPARK dataset), with a live Streamlit demo and transparent reporting of performance and limitations.

Independent Clinical Data Scientist & Bioinformatics Analyst

March 2016 – Present

Remote — serving academic medical centres, hospital research departments, and clinical investigators across Nigeria

- Built end-to-end predictive models and ML pipelines on clinical datasets spanning epidemiology, endocrinology, radiology, and internal medicine, including a logistic regression mortality model (hold-out ROC AUC 0.974).
- Developed LLM and RAG-based biomedical research applications, including a clinically grounded hMPV question-answering chatbot; built interactive analytical dashboards using Streamlit and Python.
- Created reproducible bioinformatics workflows for somatic variant calling, TP53 mutation profiling, protein characterisation, and TCGA cancer transcriptomic and survival analysis.
- Cleaned, validated, and restructured raw clinical datasets, and authored step-by-step results write-ups with traceable outputs for dissertations and manuscripts.
- Collaborated with multidisciplinary clinical teams on research design, data analysis, and manuscript preparation, contributing to 7 peer-reviewed publications.

Invited Resource Person — Biostatistics & Research Methodology

2018 – Present

Lagos University Teaching Hospital (LUTH) & Lagos State University Teaching Hospital (LASUTH), Lagos, Nigeria

- Designed and delivered biostatistics and research methodology training for 150+ postgraduate medical candidates preparing for WACP, WACS, and NPMCN fellowship examinations.
- Facilitated capacity-building sessions in-person and remotely covering study design, hypothesis testing, sample size calculation, and statistical interpretation.

EDUCATION

MSc Cell and Molecular Biology — University of Lagos, Nigeria

Expected 2027

MSc Public Health (Epidemiology) — University of Lagos, Nigeria

Coursework completed, 2021

B.Sc. Microbiology — University of Lagos, Nigeria

2016

KEY PROJECTS

Somatic WGS Pipeline — HCC1395 Tumour–Normal, SEQC2 Benchmark | github.com/Ceefour18/Breast_cancer_WGS

- Modular 7-stage tumour–normal WGS pipeline (FastQC → BWA-MEM → BQSR → Mutect2 → SnpEff) on 16 vCPU AWS, validated against the SEQC2 v1.2 gold-standard truth set.
- Debugged FASTQ read-name mismatches and cut runtime ~3× (48 h → 14 h) by migrating to Colab Pro+; each stage independently re-runnable and auditable.

Somatic WES Pipeline — HCC1143 Breast Cancer Cell Line | github.com/Ceefour18/somatic-wes-breast-cancer-gatk4

- Docker-based GATK4 Best Practices pipeline targeting chr17 (TP53, BRCA1, ERBB2): 99.47% reads mapped, 1.11% contamination; FilterMutectCalls reduced 182 raw calls to 27 PASS variants.
- Recovered the clonal TP53 p.Arg248Gln driver (VAF 98.6%); C>T transitions (52%) consistent with SBS1/SBS5 ageing signatures. Tools: BWA-MEM, GATK4, SAMtools, SnpEff, Docker, Python.

TCGA-BRCA RNA-seq — PAM50 Basal-like vs Luminal A | github.com/Ceefour18/tcga-brca-rnaseq

- 752 tumours via GDC/TCGAbiolinks; DESeq2 + apeglm identified 5,346 DEGs; GO/KEGG and GSEA enrichment; Kaplan–Meier and Cox survival modelling.

scRNA-seq Breast Cancer Tumour Microenvironment (GSE176078) | github.com/Ceefour18/scRNA_BreastCA

- Docker-containerised Seurat v5 pipeline (Harmony, scDbfFinder, SingleR, sccomp, pseudobulk DESeq2) across 9 patients and 3 subtypes; fibroblasts elevated in TNBC (FDR 0.011); 9 T cell subclusters resolved.

TP53 Cancer Genomics & In Silico Protein Characterisation | github.com/Ceefour18/tp53-cancer-genomics

- Processed 2,319 ClinVar variants (335 pathogenic); modelled five TP53 hotspot mutations with BLAST (99.46% identity) and functional domain annotation; MEGA phylogenetic analysis. Tools: Python, Biopython, R.

SELECTED PUBLICATIONS (7 PEER-REVIEWED)

- Association between Social Support and Glycemic Control among Older Type 2 Diabetes Patients. Nigerian Journal of Clinical Medicine (2024)
- Physicians' Knowledge and Barriers in Sepsis Diagnosis and Management. Journal of Clinical Sciences (2023)
- Prevalence and Determinants of Peripheral Arterial Disease in Children with Nephrotic Syndrome. PLOS ONE (2022)

Full publication list: github.com/Ceefour18/Peer-reviewed-Publication

KEY CERTIFICATIONS

- Use nf-core (Nextflow) — Seqera, Nextflow Training Week 2026
- FAS x CHIP Lecture Series 2026: AI in Medicine — Future African Scientist x Boston Children's Hospital
- GWAS (Genome-Wide Association Studies) — ISGlobal, Barcelona Institute for Global Health
- Biomedical Data Science — ISGlobal
- Research Data Management — H3Africa Consortium
- Reproducible Research — World Bank DIME
- AWS Cloud Practitioner Essentials
- IBM AI Foundations
- Generative AI — Sail Innovation Lab
- Nigerian National Code for Health Research Ethics — CITI Program (2026)
- TCPS 2: CORE 2022, Research Ethics — Panel on Research Ethics, Canada (2026)